

Global Retail Sales Dashboard

1 Project Overview

You are working as a **Data Analyst** at GlobalSuper Sales, a global retail company. The company sells **Furniture, Office Supplies, and Technology** products across multiple regions and markets.

Business Problem:

-  Profits vary across regions
-  Discounts sometimes reduce profits
-  Some orders are returned, affecting revenue
-  Sales targets differ by region

Objective:

Analyze sales, profit, returns, and regional performance using Power BI to support **data-driven decisions**.

2 Datasets Provided

Global Sales Dataset: -  GlobalSuperstore Data.xlsx

Orders Table (Fact)

Columns:

Order ID | Order Date | Ship Date | Ship Mode | Customer ID | Customer Name | Segment | Market | Region | Product ID | Category |

Sub-Category | Product Name | Sales | Quantity | Discount | Profit |
Shipping Cost | Order Priority

Purpose: Contains **all sales transactions**.

Returns Table (Dimension)

Columns:

Order ID | Returned (Yes/No) | Market

Purpose: Identify **returned orders**.

People Table (Dimension)

Columns:

Person | Region

Purpose: Assign **regional sales managers**.

3 Data Cleaning & Transformation

Steps for Students:

1. Standardize **dates** (Order Date, Ship Date)
2. Remove **currency symbols** from Sales and Profit → convert to numeric
3. Handle **missing values** (e.g., postal codes)
4. Remove **duplicate orders**
5. Create **simple calculated columns**:
 - Profit Margin (optional)
 - Returned Flag (Yes/No or 1/0)

4 Data Model

Relationships:

- Orders ↔ Returns → Order ID
- Orders ↔ People → Region

Structure:

- **Fact Table:** Orders
- **Dimension Tables:** Returns, People

5 DAX Measures

Create **basic DAX measures** using **SUM, AVERAGE, DIVIDE, COUNTROWS**:

1. Total Sales
2. Total Profit
3. Total Quantity
4. Profit Margin %
5. Return Rate %
6. Average Shipping Cost
7. Top 10 Products by Sales

6 Dashboard KPIs & Visuals

KPI Cards:

- Total Sales
- Total Profit
- Profit Margin %
- Return Rate %

Charts / Visuals:

- Line chart → Monthly Sales Trend
- Bar chart → Sales by Region
- Pie/Donut chart → Segment Contribution
- Bar chart → Top 10 Products by Sales
- KPI/Gauge → Sales vs Target

Interactive Features:

- Slicers: Year, Region, Segment, Category
- Drill-through: Region → Product → Customer
- Conditional formatting: Negative profits in red

7 Business Questions to Explore

1. Which regions have **highest & lowest profits**?
2. Are **discounts reducing profits** in some orders?
3. Which segments (Consumer, Corporate, Home Office) are **most profitable**?
4. How many orders are **returned per region**?
5. Which products generate the **most revenue**?
6. Which shipping modes are **most cost-effective**?
7. Are **regional sales managers achieving balanced performance**?

8. Deliverables

- Cleaned **Power BI data model**
- Simple **DAX measures** answering KPIs
- **Interactive dashboard** with KPIs, charts, filters
- **1–2 page Business Insights Report** with recommendations